

➤ Minimal FD Set

1. USER

User_ID $\rightarrow$ Name

User_ID $\rightarrow$ Nationality

User_ID $\rightarrow$ DOB

User_ID $\rightarrow$ Password

2. USER_PHONE

(No non-trivial FDs — relation has only key attributes)

3. USER_EMAIL

(No non-trivial FDs — relation has only key attributes)

4. JUDGE

Judge_ID $\rightarrow$ User_ID

Judge_ID $\rightarrow$ Working_status

Judge_ID $\rightarrow$ Practising_since

Judge_ID $\rightarrow$ Working_here_since

User_ID $\rightarrow$ Judge_ID

5. JUDGE_SPECIALISATION

(No non-trivial FDs — relation has only key attributes)

6. LAWYER

Bar_council_number $\rightarrow$ User_ID

Bar_council_number $\rightarrow$ Firm

Bar_council_number $\rightarrow$ Working_status

User_ID $\rightarrow$ Bar_council_number

7. LAWYER_SPECIALISATION

(No non-trivial FDs — relation has only key attributes)

8. ADMIN

Admin_ID → User_ID

Admin_ID → Post

Admin_ID → Working_here_since

User_ID → Admin_ID

9. LITIGANT

User_ID → Litigant_ID

Litigant_ID → Government_ID

Litigant_ID → Type

Litigant_ID → ID_number

10. CASE

Case_ID → CNR

Case_ID → Case_title

Case_ID → Case_type

Case_ID → Filing_date

Case_ID → Current_status

Case_ID → Description

11. CAUSE_LIST

List_ID → Courtroom_no

List_ID → Date

List_ID → Time

12. CASE_HISTORY (*Weak Entity*)

(Case_ID, Court_ID, Date) → Verdict

(Case_ID, Court_ID, Date) → Court_type

13. INTERLOCUTORY_APPLICATIONS

IA_ID → Case_ID

IA_ID → Application_type

IA_ID → Status

IA_ID → Date

14. HEARING

Hearing_ID $\rightarrow$ Case_ID
Hearing_ID $\rightarrow$ Next_date
Hearing_ID $\rightarrow$ Judge_ID
Hearing_ID $\rightarrow$ Verdict
Hearing_ID $\rightarrow$ Prosecutor_ID
Hearing_ID $\rightarrow$ Purpose
Hearing_ID $\rightarrow$ Defense_lawyer_ID
Hearing_ID $\rightarrow$ List_ID

15. EVIDENCE

Evidence_ID $\rightarrow$ Case_ID
Evidence_ID $\rightarrow$ Type
Evidence_ID $\rightarrow$ Submitted_by
Evidence_ID $\rightarrow$ Uploaded_by
Evidence_ID $\rightarrow$ Address
Evidence_ID $\rightarrow$ Submit_date

16. WITNESS

Witness_ID $\rightarrow$ Name
Witness_ID $\rightarrow$ Address
Witness_ID $\rightarrow$ Gender
Witness_ID $\rightarrow$ Witness_type

17. WITNESS_PHONE

(No non-trivial FDs — relation has only key attributes)

18. WITNESS_EMAIL

(No non-trivial FDs — relation has only key attributes)

19. PAYMENT (*Weak Entity*)

(Case_ID, Payment_no) → Payer_name
(Case_ID, Payment_no) → Amount
(Case_ID, Payment_no) → Method
(Case_ID, Payment_no) → Date
(Case_ID, Payment_no) → Status
(Case_ID, Payment_no) → Purpose

20. E_FILING

Diary_number → Document_type
Diary_number → Petitioner_ID
Diary_number → Submit_date
Diary_number → Status
Diary_number → Document_title
Diary_number → Petitioner_lawyer_ID
Diary_number → Admin_ID
Case_ID → Diary_number

21. CASE_DOCUMENT

document_id → case_id
document_id → document_type
document_id → filed_by
document_id → filed_date
document_id → hearing_id
document_id → document_address

22. FIR

FIR_number → police_station
FIR_number → district
FIR_number → date_of_filing
FIR_number → sections_invoked
FIR_number → accused_name
FIR_number → complainant_name
FIR_number → case_id

23. HAS_WITNESS

(No non-trivial FDs — relation has only key attributes)

24. CASE_JUDGES

(No non-trivial FDs — relation has only key attributes)

25. CASE_LITIGANTS

(No non-trivial FDs — relation has only key attributes)

26. CASE_LAWYERS

(No non-trivial FDs — relation has only key attributes)

